GreenLens AI | Green Customer Insights Platform

M1: Approach | Lloyds Problem Statement 1 | High-level concept and testing plan

1. Problem statement and our understanding

Banks have transaction histories, existing product relationships and digital-engagement data that can reveal opportunities for customers to adopt more sustainable financial products and behaviours. The challenge is to transform scattered records into relevant, explainable green opportunities, estimate environmental impact using defensible methods, and serve both customers and banking teams. A recommendation must not be confused with credit eligibility or a guaranteed carbon reduction.

The four mandatory opportunity families are EV Finance, Green Home Loan, Solar Finance and Paperless Banking. Every transaction must map to exactly one of eight sustainability categories: Fuel, Electricity, Gas, Public Transport, Home Improvement, Automotive, Utilities and Other. The platform must also expose propensity, confidence/reliability, reason codes, supporting evidence and data timestamps.

2. Proposed idea and core goals

GreenLens AI is a two-sided sustainability intelligence application. Its customer view explains which greener choices may be relevant and why; its business view identifies promising segments, product opportunities and potential aggregate impact. It prioritizes transparency, practical actions and reproducibility over unsupported claims.

• Customer 360: merge profiles, transactions, banking products, assets and engagement patterns into a single customer-level record.

• Opportunity detection: use trained ML models to estimate take-up propensity for each of the four products, while explicit business rules screen out inappropriate recommendations.

• Quantified impact: calculate indicative environmental effects with a deterministic, documented carbon/resource engine, reporting unavailable results where inputs are inadequate.

• Explainability and usability: provide evidence-backed natural-language insights and dashboards for customers and relationship teams.

3. Data approach

We will build a reproducible synthetic customer dataset with linked customer, transaction, product, behaviour and asset tables. Synthetic take-up outcomes will follow documented probabilistic relationships rather than arbitrary labels. Suitable public data may inform reference patterns (for example, historical electricity consumption), but will not be represented as banking records. Data provenance, generated labels, assumptions, missing values and confidentiality safeguards will be documented.

4. Scope of the working prototype

Customer experience	Business experience
Customer lookup; categorized spending; product opportunities; evidence and action prompts.	Portfolio summary; opportunities by product; segment trends; customer drill-down.

5. Proposed pipeline and system architecture

Synthetic / lawful public reference data → Data cleaning + Customer 360 → Transaction classification + feature engineering → Four ML propensity models → Eligibility/exclusion rules + documented impact calculations → Structured recommendation objects → Grounded LLM explanations → Customer and business dashboards.

Separation of responsibilities: ML estimates propensity; a rules engine enforces product constraints; deterministic formulas compute sustainability metrics; the LLM explains only the supplied structured outputs. The LLM neither invents figures nor makes financial eligibility decisions.

6. Planned modelling and implementation

• Transactions: begin with consistent category rules and benchmark a TF-IDF/text classifier where it adds value; document boundaries for Fuel vs Automotive and electricity/gas vs general Utilities.

• Propensity: use four separate supervised classifiers; benchmark Logistic Regression against a suitable tree-based model. Include preprocessing, missing-value handling, probabilistic outputs and documented reliability/calibration measures.

• Recommendation logic: attach evidence, stable reason codes, model version, data timestamp and exclusions; explicitly avoid recommending paperless enrolment to customers who are already paperless.

• Sustainability engine: compare a clearly defined baseline against an illustrative changed scenario using sourced emission factors and units. Do not interpret transaction spending as measured emissions without justified conversion assumptions.

• Delivery: FastAPI services and a responsive customer/business web interface; reproducible scripts and documented inference artefacts.

7. Planned testing and validation methodology

Area	Planned verification
Data integrity	Unique keys, valid joins, eight mutually exclusive categories, plausible synthetic relationships, label provenance and missing-data checks.
ML quality	Customer-level 70/15/15 train/validation/test separation; no leakage; baseline comparison; ROC-AUC, PR-AUC, precision/recall/F1 and probability calibration where applicable.
Impact correctness	Unit tests for formulas and emission-factor units; source tracking; unavailable estimates when input data is insufficient.
Safety and rules	Test existing EV ownership, non-homeowners, already-paperless customers, conflicting records and changed customer circumstances.
LLM + experience	Check numerical consistency and unsupported claims; validate explanations against supplied facts; demonstrate customer and portfolio workflows end-to-end.

8. Expected outcome and limitations

The expected demonstrator presents traceable, customer-specific product opportunities and portfolio-level insights. Because take-up labels are synthetic, evaluation metrics will indicate performance on simulated outcomes, not validated real-world banking accuracy. Estimated carbon impacts will be presented as scenario-based estimates rather than guaranteed reductions.